

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

---

### COURSE OUTCOME COVERED

**CO1:** Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

---

**QUESTIONS: 4**

**Total Marks:20**

### Annexure A MCQ TEST

| <i>Criteria</i>  | <b>Conceptual Understanding(3)</b> | <b>Accuracy of Answer(2.5)</b> | <b>Application of Knowledge(2)</b> | <b>Completeness of Response(1.5)</b> | <b>Clarity &amp; Presentation(1)</b> | <i>Total(10)</i> |
|------------------|------------------------------------|--------------------------------|------------------------------------|--------------------------------------|--------------------------------------|------------------|
| <i>Weightage</i> | 30%                                | 25%                            | 20%                                | 15%                                  | 10%                                  | 100%             |
| Q1               |                                    |                                |                                    |                                      |                                      |                  |
| Q2               |                                    |                                |                                    |                                      |                                      |                  |
| Q3               |                                    |                                |                                    |                                      |                                      |                  |
| Q4               |                                    |                                |                                    |                                      |                                      |                  |

---

### Code of Conduct:

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

KJOTHEESWAR REDDY

**Signature of the student:**

## RUBRICS FOR CALCULATION

| Criteria                          | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|-----------------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b>   | <b>30</b> | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>         | <b>25</b> | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b>   | <b>20</b> | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| <b>Completeness of Response</b>   | <b>15</b> | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| <b>Clarity &amp; Presentation</b> | <b>10</b> | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |



# Knowledge Discovery: From Raw Data to Smart Decisions

Name

**GUIDED BY  
BEULAH DAVID**

**PRESENTED BY  
K.JOTHEESWAR REDDY  
192425386**

# INTRODUCTION

- KNOWLEDGE DISCOVERY IN DATABASES (KDD) IS THE PROCESS OF EXTRACTING USEFUL AND MEANINGFUL KNOWLEDGE FROM LARGE VOLUMES OF DATA.
- IT TRANSFORMS RAW DATA INTO VALUABLE INSIGHTS THAT SUPPORT BETTER DECISION-MAKING.
- THE KDD PROCESS INCLUDES DATA CLEANING, DATA INTEGRATION, DATA SELECTION, DATA TRANSFORMATION, DATA MINING, PATTERN EVALUATION, AND KNOWLEDGE PRESENTATION.
- IT HELPS IDENTIFY HIDDEN PATTERNS, TRENDS, AND RELATIONSHIPS THAT ARE DIFFICULT TO DETECT THROUGH MANUAL ANALYSIS.
- KDD IS WIDELY USED IN BANKING, HEALTHCARE, E-COMMERCE, EDUCATION, AND SOCIAL MEDIA FOR DATA-DRIVEN DECISION-MAKING.

# PROJECT OBJECTIVES

- TO UNDERSTAND HOW RAW DATA IS TRANSFORMED INTO MEANINGFUL KNOWLEDGE USING THE KDD PROCESS.
- TO EXPLORE THE COMPLETE JOURNEY FROM DATA COLLECTION TO SMART DECISION-MAKING.
- TO LEARN THE SEVEN KEY STAGES OF KNOWLEDGE DISCOVERY IN DATABASES (KDD) AND THEIR IMPORTANCE.
- TO IDENTIFY HIDDEN PATTERNS, TRENDS, AND VALUABLE INSIGHTS FROM LARGE DATASETS.
- TO UNDERSTAND THE ROLE OF DATA MINING IN DISCOVERING USEFUL INFORMATION FROM DATA.

# PROBLEM STATEMENT

- THE CHALLENGE OF EXTRACTING VALUABLE KNOWLEDGE FROM MASSIVE DATA
- EVERY DAY, ORGANIZATIONS GENERATE HUGE VOLUMES OF DATA FROM ONLINE TRANSACTIONS, SOCIAL MEDIA, HEALTHCARE, BANKING, AND E-COMMERCE.
- MOST OF THIS DATA IS RAW, UNORGANIZED, AND DIFFICULT TO INTERPRET, MAKING MANUAL ANALYSIS NEARLY IMPOSSIBLE.
- IMPORTANT PATTERNS AND RELATIONSHIPS OFTEN REMAIN HIDDEN WITHIN LARGE DATASETS.
- POOR-QUALITY DATA, SUCH AS MISSING, DUPLICATE, OR INCONSISTENT RECORDS, REDUCES THE ACCURACY OF ANALYSIS.

# PROPOSED SOLUTION

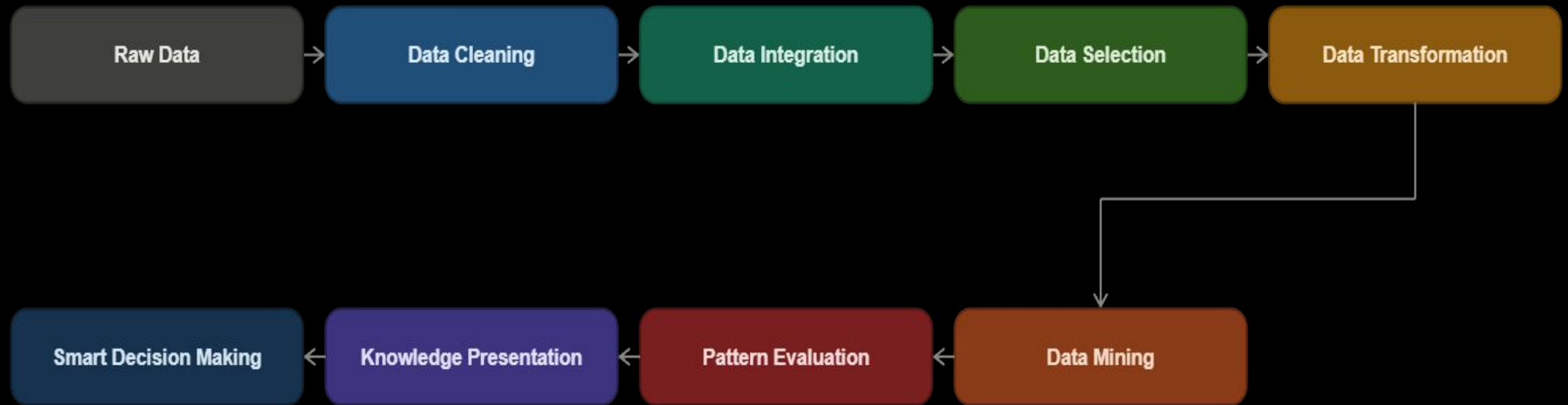
- **KNOWLEDGE DISCOVERY IN DATABASES (KDD)**
- KDD PROVIDES A STRUCTURED PROCESS TO CONVERT RAW DATA INTO VALUABLE KNOWLEDGE.

## KEY STAGES

- DATA CLEANING
- DATA INTEGRATION
- DATA SELECTION
- DATA TRANSFORMATION
- DATA MINING
- PATTERN EVALUATION
- KNOWLEDGE PRESENTATION
- **OUTCOME** : USEFUL PATTERNS AND INSIGHTS THAT SUPPORT INFORMED BUSINESS DECISIONS.



# SYSTEM DESIGN



# RESULTS

- **EXPECTED OUTCOMES**
- THE KDD PROCESS SUCCESSFULLY:
  - CONVERTS RAW DATA INTO MEANINGFUL INFORMATION.
  - IDENTIFIES HIDDEN PATTERNS AND RELATIONSHIPS.
  - SUPPORTS ACCURATE PREDICTIONS AND DECISION-MAKING.
  - IMPROVES EFFICIENCY AND BUSINESS INTELLIGENCE.
- **EXAMPLE**
  - AN ONLINE SHOPPING PLATFORM USES KDD TO RECOMMEND PRODUCTS BASED ON CUSTOMERS' PURCHASE HISTORY, INCREASING CUSTOMER SATISFACTION AND SALES.

# CONCLUSION

KNOWLEDGE DISCOVERY IN DATABASES (KDD) IS A SYSTEMATIC PROCESS THAT TRANSFORMS RAW DATA INTO MEANINGFUL AND VALUABLE KNOWLEDGE.

THE KDD PROCESS HELPS UNCOVER HIDDEN PATTERNS, TRENDS, AND RELATIONSHIPS THAT SUPPORT INFORMED DECISION-MAKING.

DATA MINING IS THE CORE STAGE OF KDD, RESPONSIBLE FOR EXTRACTING USEFUL PATTERNS FROM LARGE DATASETS.

KDD IS WIDELY USED IN BANKING, HEALTHCARE, E-COMMERCE, EDUCATION, AND BUSINESS ANALYTICS TO IMPROVE EFFICIENCY AND SOLVE REAL-WORLD PROBLEMS. AS DATA CONTINUES TO GROW,

KDD WILL REMAIN A KEY TECHNOLOGY FOR DRIVING INNOVATION AND INTELLIGENT DECISION-MAKING IN THE FUTURE.